



Hybrid Monitoring For Side-Channel Security In Edge SoCs

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MOTIVATION & OBJECTIVE

- Advanced Encryption Standard (AES) is widely used in embedded and edge systems for secure data processing.
- However, side-channel attacks based on power and timing analysis expose vulnerabilities, especially under real-time execution constraints.
- Traditional threshold-based anomaly detection methods often fail under high computational load or dynamic system behavior.
- This work proposes a hybrid monitoring approach that integrates machine learning to enable adaptive, scalable, and real-time anomaly detection without interrupting encryption.

RESEARCH GOAL & EVALUATION FOCUS

- This study evaluates the effectiveness of machine learning (ML) compared to traditional threshold-based anomaly detection for AES encryption.
- Experiments analyze performance across varying block sizes and malicious injection rates.
- Key metrics include detection accuracy, precision, and adaptability under increasing system load.

RESULTS AND ANALYSIS

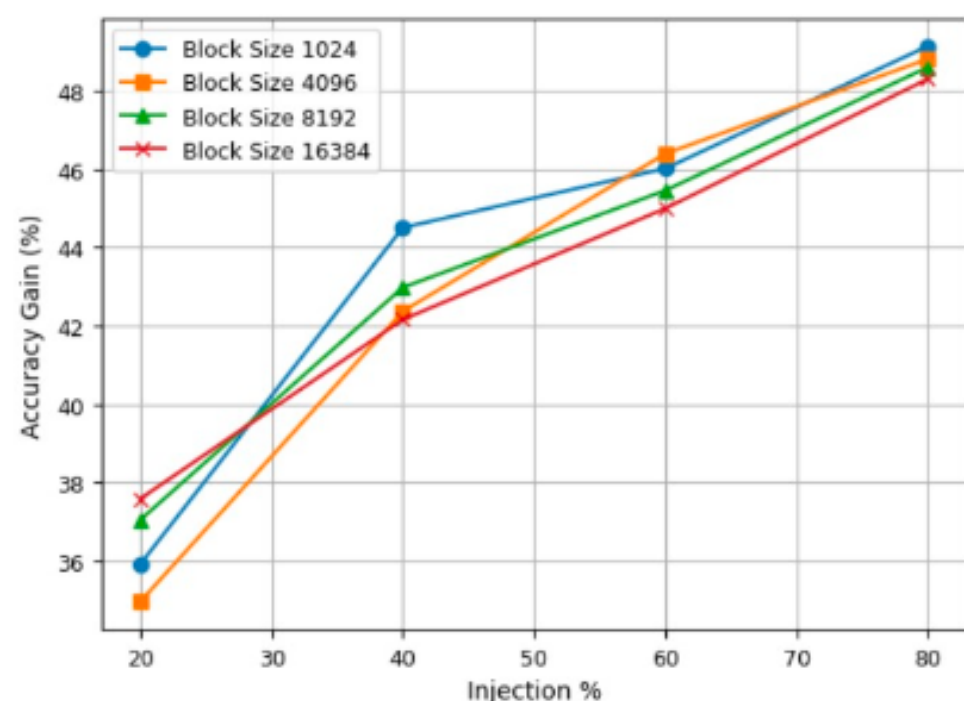


Figure 1: Accuracy Gain of ML-based Detection Across Injection Rates and Block Sizes. ML-assisted detection consistently outperforms threshold methods, with higher accuracy gains observed at larger block sizes and higher injection percentages.

EXPERIMENTAL SETUP & PIPELINE

We conducted experiments on macOS 15.7 using Python 3.13.2 and AES-128 (ECB) from PyCryptodome. Execution time per block was used to simulate power-based features. Tests covered block sizes from 1,024 to 16,384, injection rates from 10% to 80%, and 1 to 8 CPU cores. Each run included AES encryption with timed malicious injection, followed by classification using either a threshold rule (based on execution time) or a DecisionTreeClassifier trained on the timing data.

REAL-TIME DETECTION IN ACTION: THRESHOLD VS. ML

We benchmarked ML vs. threshold detection; ML showed higher accuracy, precision, and stability across all conditions.

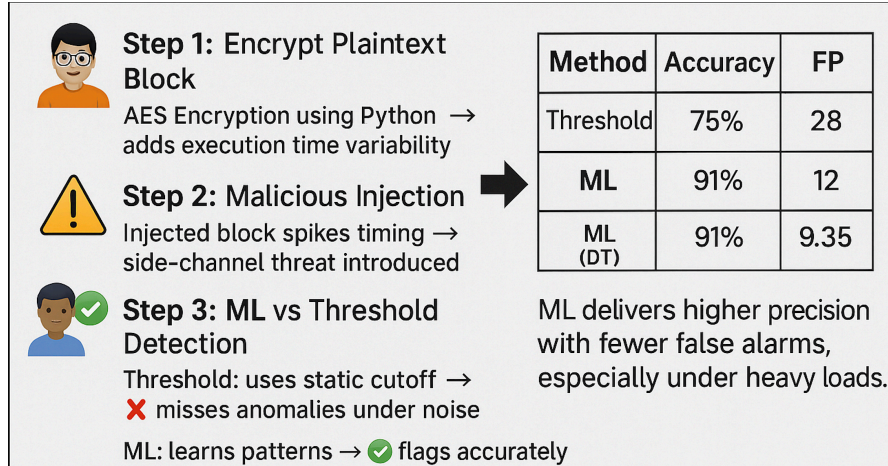


Figure 2: Real-Time Detection Workflow and Performance Comparison Between Threshold and ML Methods

- ML-based detection shows increasing accuracy as injection rates and block sizes grow, demonstrating better scalability compared to threshold-based methods.

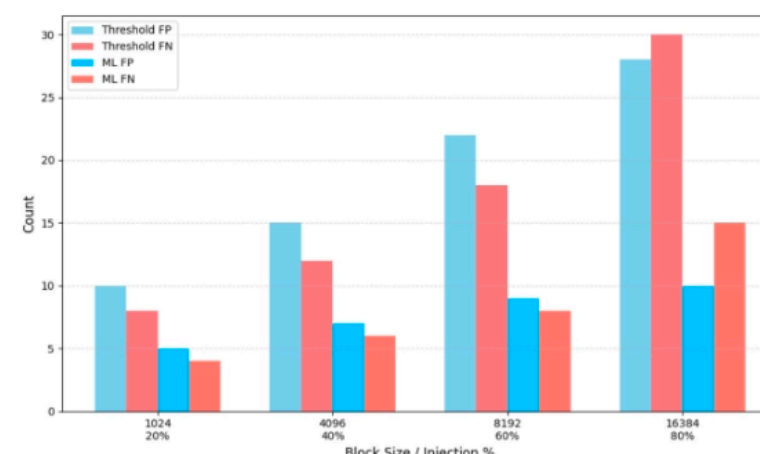


Figure 4: False Positive and False Negative Comparison Across Block Sizes and Injection Rates.

- Side-by-side comparison of real-time AES anomaly detection using threshold rules and machine learning.
- ML improves detection accuracy and significantly reduces false positives under varying conditions.

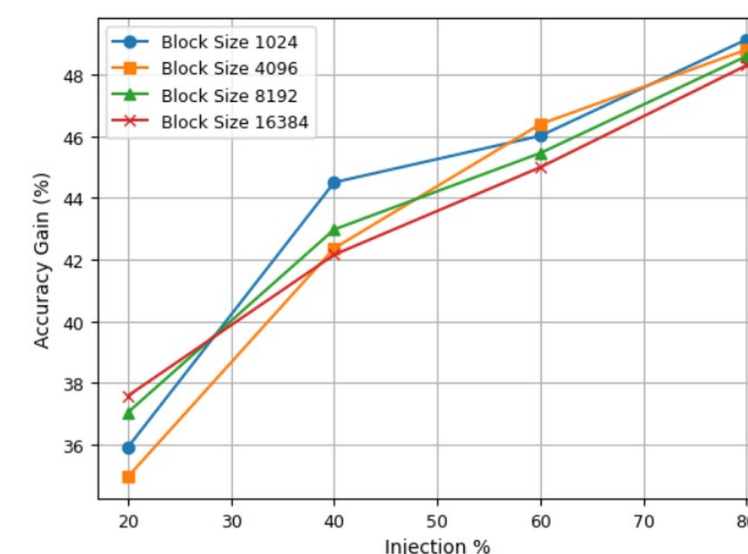
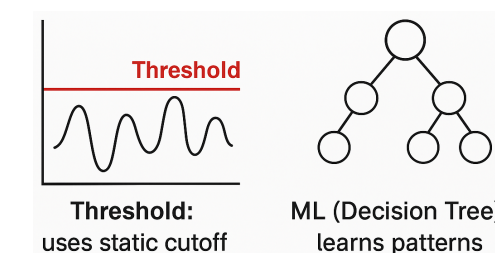


Figure 3: Accuracy Gain of ML-Based Detection Across Injection Rates and Block Sizes

- ML consistently produces fewer false positives and false negatives, especially under higher injection loads and dynamic execution conditions.

SCALABILITY & REAL-TIME SUITABILITY

- ML detection adapts to high block sizes and injection rates
- Maintains precision under real-time AES encryption
- Suitable for embedded and timing-sensitive applications



WHY ML OUTPERFORMS THRESHOLD

- Threshold detection uses fixed cutoff time → inflexible under variability
- ML (Decision Tree) learns pattern from timing behavior → adapts dynamically
- ML can capture subtle anomalies that threshold methods miss under noise/load

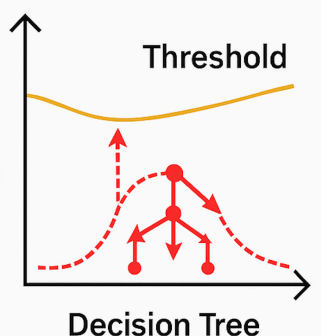


Illustration:

A fixed threshold line fails to capture nuanced changes in timing data, while ML (e.g., Decision Tree) dynamically adapts to patterns, improving detection under variable conditions.

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